

Project Workflow: Week 6

Phase: Obstacle Cars & Collision Detection

1. Objective

The goal of Week 6 was to introduce obstacle traffic and hazard/reward detection into the game. This included:

1. Implementing EnemyCar with randomized colors, speeds, and lane positions.
2. Adding AABB collision detection between the player and enemy cars.
3. Implementing GasItem collectible pickups with spin/glow animation and speed boost effect.

2. Detailed Task Breakdown

❖ Implementing EnemyCar

Created the EnemyCar class extending the abstract Car class from Week 5. Each enemy car is assigned a random color from a fixed palette and a random speed, with two distinct behaviors: cars "coming towards" the player move downward faster than the player's scroll speed, while cars moving in the same direction scroll upward more slowly. Enemy cars are spawned periodically at random lanes from both above and below the visible screen, and are removed once they scroll far enough off-screen to keep the object list from growing indefinitely.

❖ Adding AABB Collision Detection

Implemented axis-aligned bounding box (AABB) collision checking in the Car class, comparing the player's x/y/width/height against each enemy car's bounds every frame. When an overlap is detected, the game immediately transitions to a Game Over state and halts the update loop. The same intersects() logic was overloaded to check overlap between the player and GasItem pickups, reusing one consistent collision method across different object types. Cars that are safely passed (rather than collided with) are also tracked so they can be scored.

❖ Implementing GasItem Pickups

Built the GasItem class as a rotating, glowing collectible that scrolls down the road in sync with the player's speed, snapping toward the correct lane position each frame. On collision with the player, the item is removed, the score increases, and the player's speed receives a temporary boost via PlayerCar's collectGas() method, capped at a maximum speed so the boost stays balanced.

3. Deliverables

1. EnemyCar Class: Randomized obstacle traffic with two movement behaviors (approaching / departing).
2. Collision Detection: AABB-based intersects() method shared between Car-Car and Car-GasItem checks.
3. GasItem Class: Collectible pickup with spin/glow animation and a speed-boost effect on collection.
4. Updated GamePanel: Enemy spawning, gas spawning, collision handling, and scoring wired into the Week 4 game loop.

4. Tools & Resources

❖ Development Environment

1. Java JDK 8+
2. IntelliJ IDEA / VS Code

❖ Reference Materials

Java 2D Graphics2D and AffineTransform documentation (for the GasItem spin animation), AABB collision detection concepts, and object-oriented design references for method overloading (intersects(Car) vs intersects(GasItem)).